

3. Project Design Phase

ComicCraft – AI Comic Story Creator

Introduction

The design phase defines how the required ComicCraft components will work together. The design separates the web interface, API layer, AI services, image processing, PDF creation, configuration, and file utilities.

System Architecture

The user interacts with the responsive frontend. JavaScript uses the Fetch API to communicate with FastAPI. FastAPI receives and validates requests, invokes the Gemini service for structured story generation, invokes the image service for comic illustrations, and invokes the PDF service for the final document. Generated images and PDFs are stored in their respective generated folders.

Module Design

The main backend modules are `app/main.py`, `app/config.py`, and `app/schemas.py`. Service modules include `gemini_service.py`, `image_service.py`, and `pdf_service.py`. `file_utils.py` provides file-related utilities. The frontend consists of `index.html`, `style.css`, and `app.js`.

Story Data Design

Gemini output is designed as structured JSON containing a title, summary, character profile, panels, narration, dialogue, scene descriptions, and image prompts. Pydantic validation is used to validate this structured output before it is used by later services.

Image Design

Each image prompt should carry character appearance, action, environment, lighting, camera angle, mood, and art style so that generated panels contain the information needed for a coherent comic illustration.

PDF Design

The PDF contains the title, summary, panel images, narration, dialogue, and page numbers. ReportLab is used for PDF creation, while generated images are stored in `generated/images/` and PDFs in `generated/pdfs/`.

Frontend Design

The interface is designed around a comic-book style and responsive layout. It includes a creation form, Generate button, loading status, comic preview, panel cards, Regenerate Panel button, Download PDF button, and New Comic button.

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